

JACK MICHELL

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Summary

My dream career is to become a Software Engineer. I plan on learning anything that interests me even the slightest. A top career goal I have is to use these interests to gain knowledge and experience and to see how things connect and why. By gaining this experience through pure interest and internships, I plan to create software that benefits people as well as other developers. I believe even the smallest details matter when building software, and I aspire to be able to contribute in this field.

Education

Tarleton State University – 3.73 GPA

2025 – 2028

Pursuing Bachelor of Science, Computer Science (Software Engineering), Minor in Mathematics

Previous Work Experience

Student Intern - Researcher

May 2025– August 2025

Tarleton State University

Supervisor: Joseph Meier, Dept. CSEE

- Designed and implemented computer vision algorithms for autonomous landing pad detection using Python and OpenCV, comparing effectiveness of PnP, K-means clustering, and edge detection methods.
- Integrated YOLO object detection model with traditional computer vision techniques to improve detection accuracy and minimize error.
- Conducted comparative analysis of multiple detection algorithms by testing accuracy and comparing projected points.
- Collaborated on research implementation with three other teams and presented findings on detection accuracy.
- Optimized detection performance through algorithm tuning and code optimization, reducing processing time while improving detection to 60% with minimal error.

Co-Founder - Helvera LTD

August 29, 2021– October 25, 2022

Software and Game Development

- Managed and organized projects, teams, and company finances including but not limited to budgeting.
- Worked closely with individual contributors on the development of games including designing, testing, implementation, and production releases.
- Hosted productive meetings and talks with individual contributors and teams to come up with innovative solutions to problems.
- Managed finances and business for the company including budgets, contracts, advertisement plans, and partners.

Projects and Extracurriculars

President's Excellence in Research Scholars

May 2025– August 2025

Tarleton State University

Supervisor: Joseph Meier, Dept. CSEE

- Project Title: Autonomous Drone Landing Pad Detection
- Role: Researcher
- Implemented and compared multiple computer vision detection methods including Perspective-n-Point (PnP), K-means clustering, homography, and edge detection algorithms, integrating with a YOLO object detection model to minimize detection error.
- Utilized data science techniques such as statistical analysis, data preprocessing, and feature extraction to improve detection accuracy.
- Collaborated with multiple teams to develop a physical autonomous drone capable of detecting and landing on a designated moving target.

Petoi Robotics

February 2025– Present

Tarleton State University

Supervisor: Dr. Gloria Frago-Diaz, Dept. ET

- Project Title: Petoi Robotic Dogs
- Role: Team Member/Contributor
- Designed, manufactured, built, and programmed robotic animals using open source software and tools provided by Arduino.
- Implemented functionality by programming and refactoring skills for our own purpose including different tricks that can be performed by voice commands.
- Collaborated with four other students along with multiple other teams to design and build a robotic dog that was presented in a showcase.

Skills and Tools

- Skills: Technical Documentation, Collaboration, Problem-solving, Academic Research
- Operating Systems: Windows 10/11, Ubuntu/Arch Linux Distributions
- Languages: Python, C, Java, JavaScript, TypeScript
- Frameworks: React, Tailwind, Next.js
- Development Tools: Git, Neovim, IntelliJ, Command-line, Visual Studio, VSCode
- Libraries: OpenCV, scikit-learn, Matplotlib, Pandas, NumPy